

Project:	Type:
Product Code:	Date:

V022626



2" Square Downlight

SMALL APERTURE Delivers the ultimate in quiet ceiling designs

PRECISION LOCK COLLAR Eliminates over-cutting and ensures perfect cutout

ARCHITECTURE-GRADE Deep die-cast aluminum knife edge trim provide unmatched precision and exceptionally easy installation

DIMMING COMPATIBILITY GUARANTEED In-house designed integrated driver ensures seamless compatibility with every major lighting control system

INSTALLATION

Ceiling Thickness

New Construction: 1/2" up to 1 1/4"
 Remodel: 1/2" up to 1 1/4"

Ceiling Material

Drywall, Millwork, Acoustic Ceiling Tiles

TRIMS

Style

Standard Micro Flange, Flangeless, Wall Wash, Pinhole, Hyperbolic, Decorative

Finish

White, Black, Bronze, Clear Diffuse, Custom Color (RAL Colors)

Premium Metalized Finish

Brass, Copper, Graphite

LIGHT OUTPUT & DISTRIBUTION

Module

Downlight

Lumens (Power)

750 lm (9.5W), 1000 lm (12.5W), 1250 lm (14.3W TRIAC, 14.5W DALI-2)

Color Quality

True Spectrum®

Color Temperature

- 2700K

3000K

3500K
- 4000K

Warm Dim (3000K–1800K)
- Tunable White (4000K–1800K)

Beam Spread

- 20° Narrow Spot

30° Spot
- 40° Flood

50° Wide Flood
- 50°x30° Linear Spread

POWER & CONTROLS

Input Voltage

120 / 277V

Dimming

TRIAC/ELV (1%, 120V only), DALI-2 (0.1%)

RATINGS & CERTIFICATIONS

- IC (Insulation Contact) Rated ¹

ASTM E283 Certified Air Tight

Chicago Plenum

UL Listed for Wet Location ²

RoHS Compliant

NSF Listed ³

DarkSky Approved ⁴

¹ New Construction only; IC for Roll, Batt, Loose Fill and Spray Foam see Housing section for details

² Flanged Trims, Under Covered Areas Only

³ White and Black finish Only

⁴ Downlight only, CCT less than or equal to 3000K, non-decorative trims Only

WARRANTY

10 year limited warranty; 50,000 hours



PRODUCT BUILDER

SQUARE HOUSING

EXAMPLE: X2NCS

PRODUCT CODE		INSTALLATION		SHAPE	
X2				S	
X2	2" Housing	NC	New Construction	S	Square
		CP	Chicago Plenum		
		RM	Remodel ¹		
		RD	Remodel, Digital Modules (DALI/PhaseX) ²		

¹ Non-IC installation only, Flanged and Micro Flange trims options only;
Not available with DALI-2 dimming

² Only available with DALI-2 dimming

MODULE

EXAMPLE: ART2D07T27NST

PRODUCT CODE		LUMENS		COLORMETRIC		CCT ³		BEAM SPREAD		DIMMING	
ART2D				T							
ART2D	Downlight	07	750 lm	T	True Spectrum®	27	2700K	NS	Narrow Spot (20°)	T	TRIAC/ELV
		10	1000 lm			30	3000K	SP	Spot (30°)	D	DALI-2 ⁶
		12	1250 lm			35	3500K	FL	Flood (40°)		
						40	4000K	WF	Wide Flood (50°)		
						3W	Warm Dim (3000K - 1800K) ⁴	LS	Linear Spread (50°x30°)		
						T1	Tunable White (4000K - 1800K) ⁵				

³ DarkSky approved is only applicable to CCT options less than or equal to 3000K; Not compatible with decorative trims

⁴ Only available in 1000 lm; Not available in Narrow Spot beam spread

⁵ Only available in 1000 lm, DALI-2 dimming; Not available in Narrow Spot beam spread

⁶ Only available in 1000 lm for Tunable White CCT and Warm Dim or 1250 lm with static CCT; Not compatible with Standard Remodel Housing, Digital Remodel only

SQUARE TRIM

EXAMPLE: X2TSDSWHMF

PRODUCT CODE		FINISH		OPTION	
X2TSDS	2" Standard Trim	WH	White	[Blank]	(Pinhole or Hyperbolic Only)
X2TSDW	2" Wall Wash Trim ⁷	BK	Black	MF	Micro Flange
X2TSDP	2" Pinhole Trim ⁸	BZ	Bronze	FL	Flangeless ¹¹
		CW	Clear Diffuse, White Flange ⁹	DO	Decorative: Open ¹² (Coming Soon)
		CC	Custom (RAL Colors)	DF	Decorative: Float ¹² (Coming Soon)
		PREMIUM METALIZED FINISH			
		BR	Brass ¹⁰		
		CO	Copper ¹⁰		
		GR	Graphite ¹⁰		

⁷ Recommended with Wide Flood beam spread

⁸ Only available in White or Black finish; Not available in Micro Flange option

⁹ Only available in Micro Flange (Standard or Wall Wash)

¹⁰ Only available in Standard style

¹¹ Not available in Clear Diffuse; Requires Flangeless Mud or Wood Kit; Not compatible with Remodel housing

¹² Only available in 750 or 1000 lm, Standard style, White finish; Not available in Custom Color finish



PRODUCT BUILDER

OPTICS

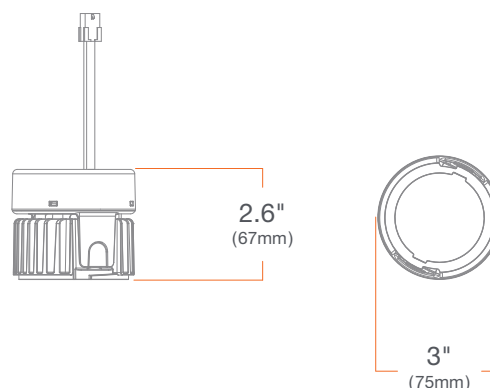
DXOP	Bundled Downlight Optics Kit <i>(Includes Narrow Spot, Spot, Flood, Wide Flood and Linear Spread)</i>
DXNS	Narrow Spot (20°) Optic
DXSP	Spot (30°) Optic
DXFL	Flood (40°) Optic
DXWF	Wide Flood (50°) Optic
DXLS	Linear Spread (50°x30°) Optic
DXHD	High Diffusion Front Lens <i>(Can be paired with all beam spreads to provide extra diffusion; Only available as an optical accessory)</i>
DXCD	Clear Front Lens <i>(Can be paired with all beam spreads to decrease diffusion and define edges; Only available as an optical accessory)</i>

ACCESSORIES

X2KSMUD	Flangeless Mud Kit <i>(Required for New Construction installation of Flangeless Trim in drywall ceiling)</i>
X2KSWOOD	Wood Kit <i>(Required for New Construction or Remodel installation in wood ceiling)</i>
X2KSTEMPLATE	Router Template <i>(Recommended for routing 3 1/4" [82mm] cutout in wood ceiling)</i>
X2KSRETRO6	6" to 2" Retrofit Conversion Kit ¹³

¹³ For more detailed Information, please refer to [Retrofit Conversion Kits spec sheet](#)

MODULE



SUMMARY

	STATIC CCT	WARM DIM	TUNABLE WHITE
CCT	2700K, 3000K, 3500K, 4000K	3000K - 1800K	4000K - 1800K
LUMENS (POWER)	750 lm (9.5W), 1000 lm (12.5W), 1250 lm (14.3W TRIAC, 14.5W DALI-2)	1000 lm (12.5W)	1000 lm (14.5W)
EFFICACY	79 lm/W, 80 lm/W, 87 lm/W, 86 lm/W, 105 lm/W	80 lm/W	69 lm/W
COLOR QUALITY	True Spectrum® 98 CRI, 2-step SDCM	True Spectrum® 93 CRI, 2-step SDCM	True Spectrum® 93 CRI, 3-step SDCM
BEAM SPREAD	Narrow Spot (20°), Spot (30°), Flood (40°), Wide Flood (50°), Linear Spread (50°x30°)	Spot (30°), Flood (40°), Wide Flood (50°), Linear Spread (50°x30°)	Spot (30°), Flood (40°), Wide Flood (50°), Linear Spread (50°x30°)
DIMMING	TRIAC/ELV, DALI-2	TRIAC/ELV, DALI-2	DALI-2
LED CHIP	Cree Pro9	Bridgelux	Bridgelux
LISTINGS	ENERGY STAR® qualified ¹ , California Title 24 2019–2022 JA8-E Compliant, UL Listed for Wet Location ² , UL Certified US-CA, Declare, RoHS compliant, NSF/ANSI 2 Listed ³ , Dark Sky Sompliant ⁴		

¹ Refer to ENERGY STAR Certified light fixtures database

² With Flanged trims, under covered areas only

³ In accordance to NSF/ANSI 2 for Splash Zone having designated trim finishes

⁴ Only applicable to CCT options less than or equal to 3000K and not compatible with decorative trims

ADDITIONAL

INPUT VOLTAGE: 120/277V, 50/60Hz

MAX INPUT CURRENT (120V): 0.13 amps

MAX INPUT CURRENT (277V): 0.06 amps

CONNECTOR: 6-pin Molex

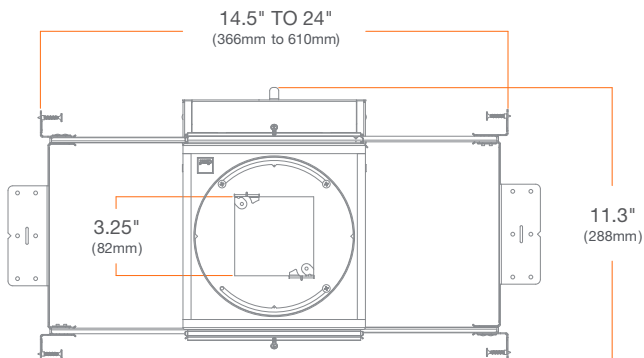
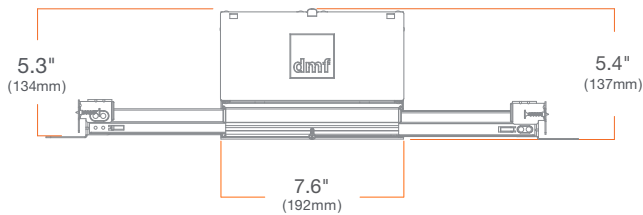
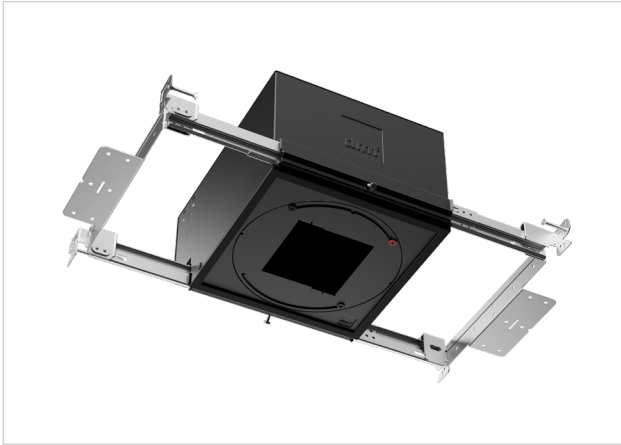
POWER FACTOR: Greater than 0.9

TOTAL HARMONIC DISTORTION: Less than 20%

AMBIENT OPERATING TEMPERATURE: -20°C to 40°C

WARRANTY: 10 year limited warranty; 50,000 hours at 70% lumen maintenance

HOUSING



New Construction **X2NCS** Chicago Plenum **X2CPS**

SUMMARY

JUNCTION BOX: Equipped with 10 knockouts: four 1/2" trade size knockouts, two 3/4" trade size knockouts, and four romex knockouts. Approved for 8 (four in, four out) #12 AWG 70°C.

MOUNTING: Adjustable Bar Hangers for 14 1/2" to 24" joist spacing, when bar hangers are broken down, up to 10.5" of joist spacing. Compatible with traditional joists, Armstrong TM ceiling, Hat Channel and T-Bar. Vertical adjustment (1/2"-1 1/4") through installation of the square collar. Front plate can be adjusted by 90°.

CEILING: 1/2" up to 1 1/4"

CUTOUT: 3 3/4" (82mm) square opening

LISTINGS: Chicago Plenum, ASTM E283 certified Air Tight, IC (Insulation Contact) up to R-32 for Roll, Batt, and Loose Fill, up to R-60 Spray Foam with standard trims, up to R-32 Spray Foam with other trims or hex louver.

HOUSING



Remodel

X2RMS

SUMMARY

JUNCTION BOX: Equipped with four knockouts: two 1/2" trade size knockouts and two romex knockouts. Approved for 8 (four in, four out) #12 AWG 70°C.

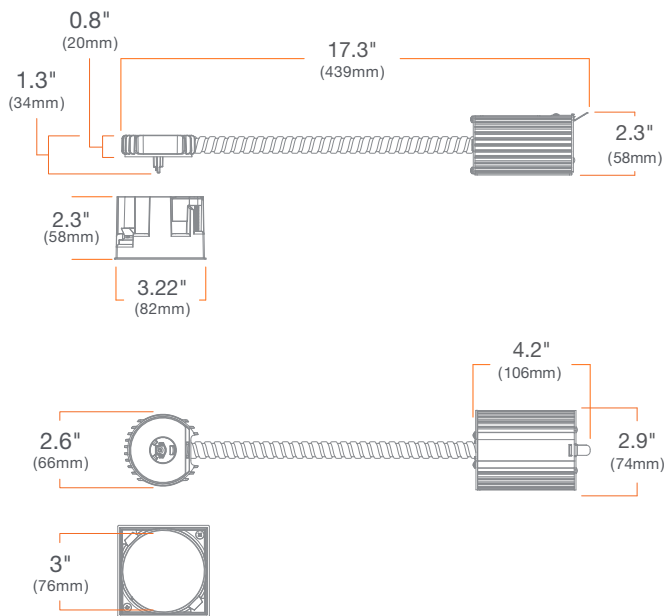
CEILING:

Drywall: 1/2" - 1 1/4" (12mm-32mm)

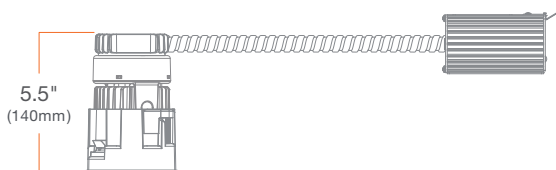
Wood Ceiling: 1/2" - 1" (12mm-25mm)

Acoustic Ceiling Tiles: 5/8" - 1 1/4" (16mm-32mm)

CUTOUT: 3 1/4" (82mm) round opening



Total Height in Plenum



HOUSING



Remodel, Digital Module (DALI/PhaseX)

X2RDS

SUMMARY

JUNCTION BOX: Equipped with four knockouts: two 1/2" trade size knockouts and two romex knockouts. Approved for 8 (four in, four out) #12 AWG 70°C.

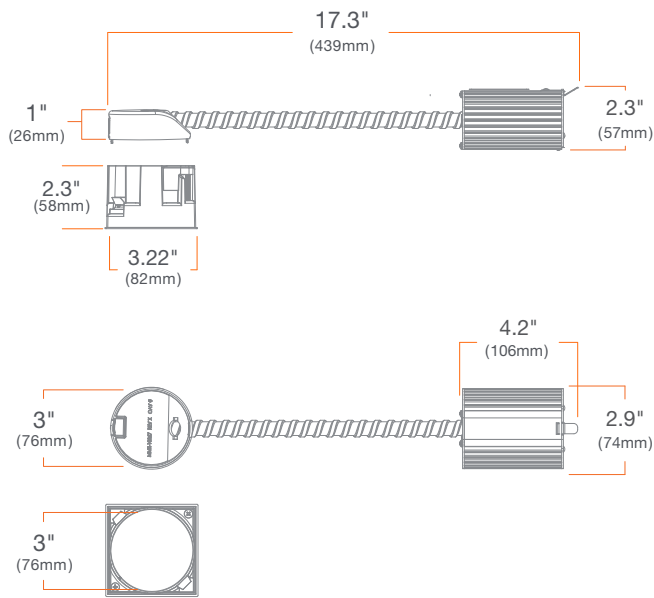
CEILING:

Drywall: 1/2" - 1 1/4" (12mm-32mm)

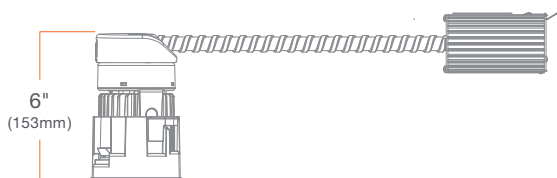
Wood Ceiling: 1/2" - 1" (12mm-25mm)

Acoustic Ceiling Tiles: 5/8" - 1 1/4" (16mm-32mm)

CUTOUT: 3 1/4" (82mm) round opening



Total Height in Plenum



TRIMS



SUMMARY

CONSTRUCTION: Die-cast aluminum

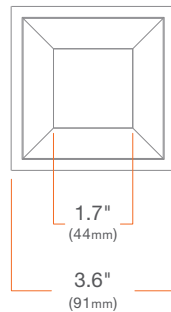
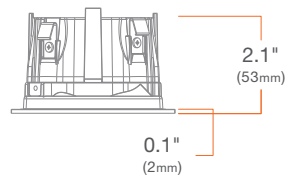
FINISH: Wet paint

DECORATIVE: Acrylic diffuser

INSTALLATION: Precision Lock Collar included with Micro Flange Trim and Flangeless Mud or Wood Kit

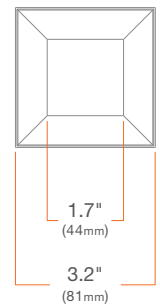
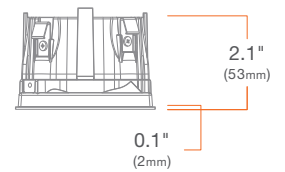
Standard Micro Flange

X2TSDSWHMF



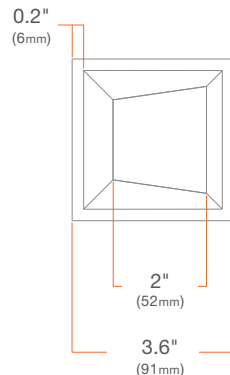
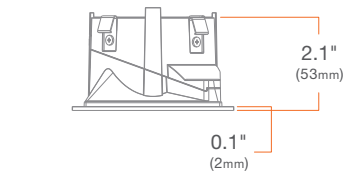
Standard Flangeless

X2TSDSWHFL



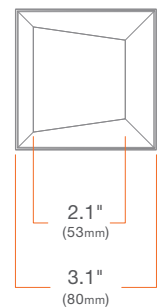
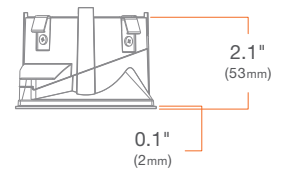
Wall Wash Micro Flange

X2TSDWWHMF



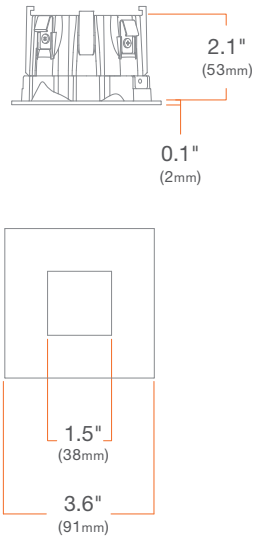
Wall Wash Flangeless

X2TSDWWHFL



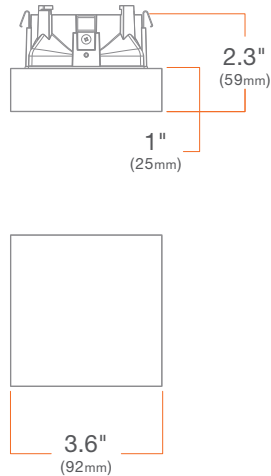
Pinhole

X2TSDPWH



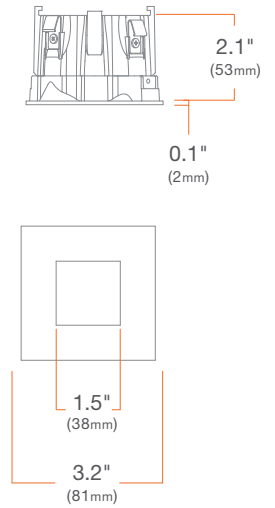
Decorative: Float

X2TSDSWHDF



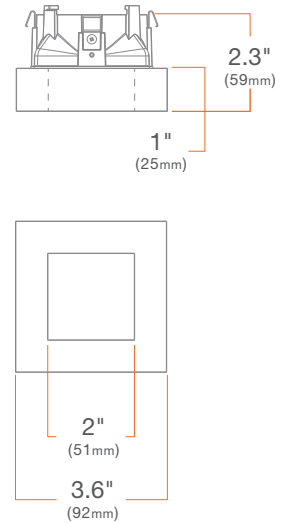
Pinhole Flangeless

X2TSDPWHFL



Decorative: Open

X2TSDSWHDO



ACCESSORIES

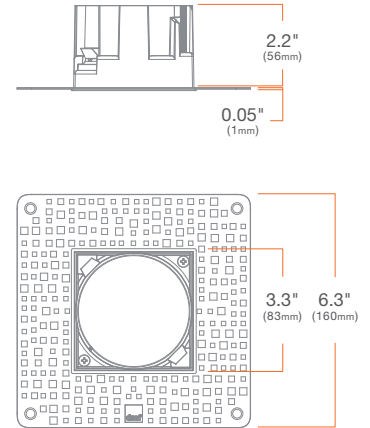


Flangeless Mud Kit

X2KSMUD

SUMMARY: Flangeless Mud Kit (Required for New Construction installation of Flangeless Trim in drywall ceiling)

CEILING: 1/2" up to 1 1/4"

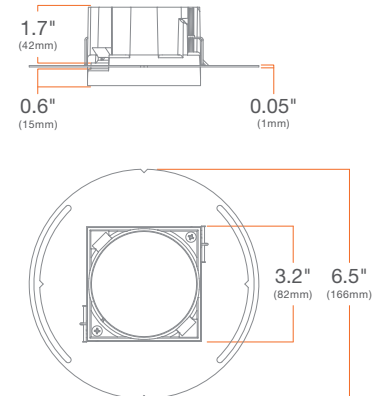


Wood Kit

X2KSWOOD

SUMMARY: Wood Kit (Required for New Construction installation of all trims in wood ceiling. Must be mounted to wood prior to ceiling installation.)

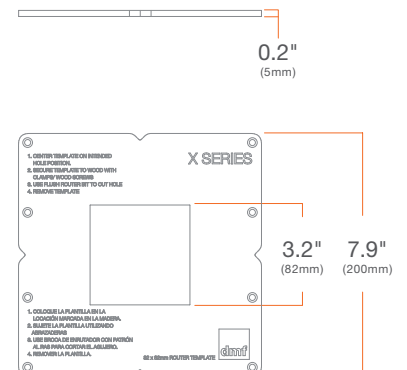
CEILING: 1/2" up to 1"



Router Template

X2KSTEMPLATE

SUMMARY: Router Template (Recommended for routing 3 1/4" [82mm] cutout in wood ceiling)



ACCESSORIES

For more detailed Information, please refer to [Retrofit Conversion Kits spec sheet](#)

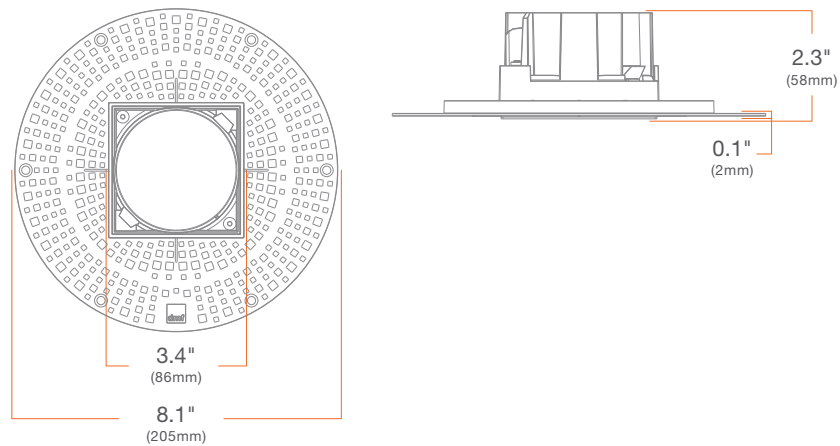


Retrofit Kit - 6" to 2" Square Conversion

X2KSRETRO6

SUMMARY: Retrofit Conversion Kit (Used to convert existing 6" fixture to a 2" fixture)

CEILING: Can be installed in any ceiling thickness

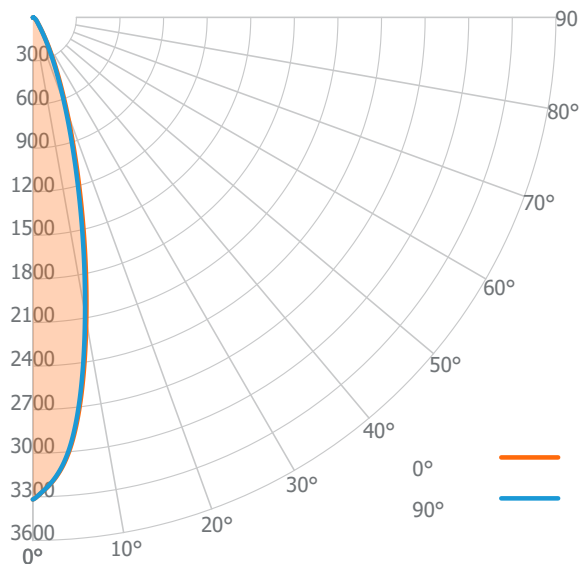


PHOTOMETRY

1000 lm, 3000K, Narrow Spot **ART2D-10T30NST**

2" Standard Micro Flange Trim, White **X2TSDSWHMF**

Candela Distribution



Luminous Intensity

	0°
0°	3318
5°	2972
10°	2119
15°	1258
20°	682
25°	360
30°	183
35°	97
40°	59
45°	42
50°	31
55°	23
60°	16
65°	11
70°	6
75°	3
80°	2
85°	1
90°	1

Zonal Lumen Summary

Zone	Lumens	Luminaire %
0° - 30°	708	87%
0° - 40°	760	93%
0° - 60°	803	98%
0° - 90°	816	100%
0° - 180°	816	100%

Illuminance Chart

Distance from LED	Foot Candles	Diameter
3 ft	368.7 fc	1.3 ft
6 ft	92.2 fc	2.7 ft
9 ft	41.0 fc	4.0 ft
12 ft	23.0 fc	5.3 ft

Input Watts: **12.3 W**

Lumens: **816 lm**

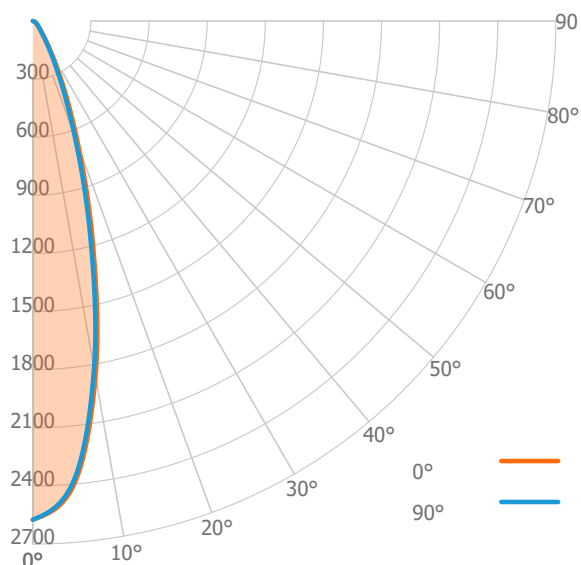
Efficacy: **66 lm/W**

Beam Angle: **25° x 24°**

1000 lm, 3000K, Spot **ART2D-10T30SPT**

2" Standard Micro Flange Trim, White **X2TSDSWHMF**

Candela Distribution



Luminous Intensity

	0°
0°	2574
5°	2408
10°	1863
15°	1203
20°	690
25°	374
30°	198
35°	106
40°	64
45°	44
50°	33
55°	24
60°	17
65°	11
70°	6
75°	3
80°	2
85°	1
90°	0

Zonal Lumen Summary

Zone	Lumens	Luminaire %
0° - 30°	652	85%
0° - 40°	709	92%
0° - 60°	754	98%
0° - 90°	767	100%
0° - 180°	767	100%

Illuminance Chart

Distance from LED	Foot Candles	Diameter
3 ft	286.0 fc	1.5 ft
6 ft	71.5 fc	3.1 ft
9 ft	31.8 fc	4.6 ft
12 ft	17.9 fc	6.1 ft

Input Watts: **12.0 W**

Lumens: **767 lm**

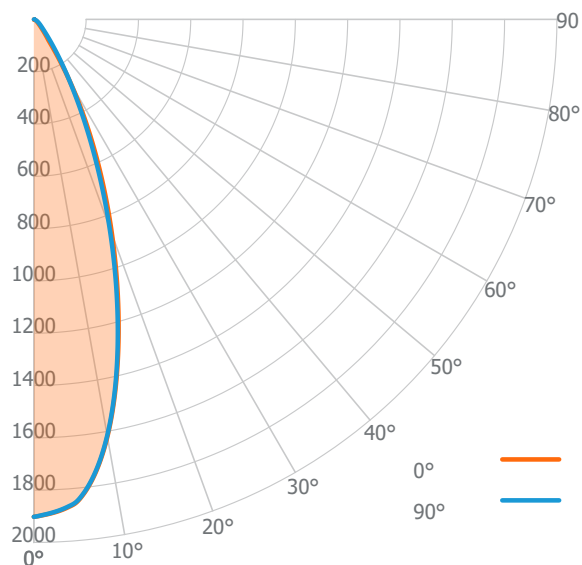
Efficacy: **64 lm/W**

Beam Angle: **29° x 28°**

PHOTOMETRY

1000 lm, 3000K, Flood **ART2D-10T30FLT**2" Standard Micro Flange Trim, White **X2TSDSWHMF**

Candela Distribution



Luminous Intensity

	0°
0°	1902
5°	1855
10°	1620
15°	1252
20°	874
25°	546
30°	283
35°	99
40°	52
45°	30
50°	20
55°	14
60°	10
65°	7
70°	4
75°	2
80°	2
85°	1
90°	1

Zonal Lumen Summary

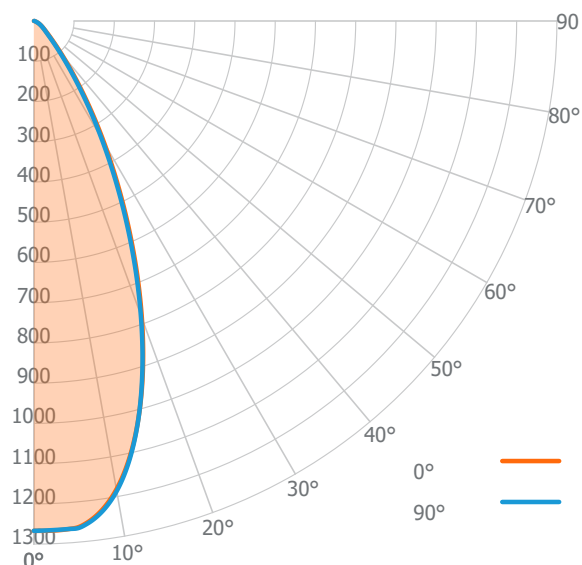
Zone	Lumens	Luminaire %
0° - 30°	693	85%
0° - 40°	763	94%
0° - 60°	801	99%
0° - 90°	813	100%
0° - 180°	813	100%

Illuminance Chart

Distance from LED	Foot Candles	Diameter
3 ft	211.3 fc	2.1 ft
6 ft	52.8 fc	4.1 ft
9 ft	23.5 fc	6.2 ft
12 ft	13.2 fc	8.2 ft

Input Watts: **12.0 W**Lumens: **813 lm**Efficacy: **68 lm/W**Beam Angle: **38° x 37°**1000 lm, 3000K, Wide Flood **ART2D-10T30WFT**2" Standard Micro Flange Trim, White **X2TSDSWHMF**

Candela Distribution



Luminous Intensity

	0°
0°	1267
5°	1263
10°	1182
15°	1011
20°	785
25°	549
30°	349
35°	199
40°	107
45°	60
50°	38
55°	27
60°	19
65°	12
70°	7
75°	4
80°	3
85°	1
90°	0

Zonal Lumen Summary

Zone	Lumens	Luminaire %
0° - 30°	593	77%
0° - 40°	697	91%
0° - 60°	755	98%
0° - 90°	770	100%
0° - 180°	770	100%

Illuminance Chart

Distance from LED	Foot Candles	Diameter
3 ft	140.8 fc	2.6 ft
6 ft	35.2 fc	5.1 ft
9 ft	15.6 fc	7.7 ft
12 ft	8.8 fc	10.3 ft

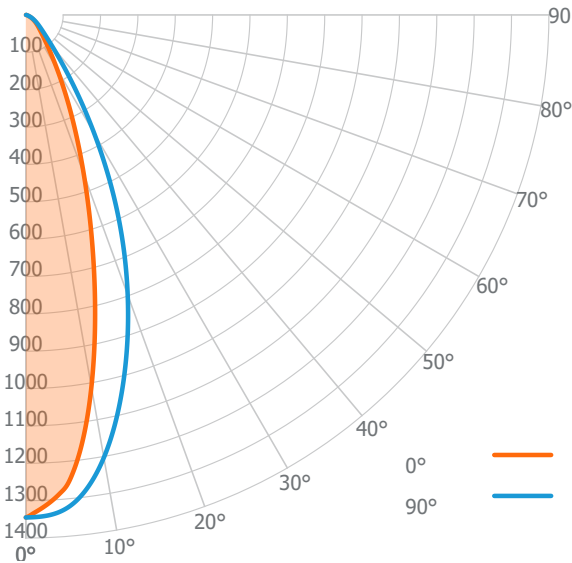
Input Watts: **12.0 W**Lumens: **770 lm**Efficacy: **64 lm/W**Beam Angle: **46° x 46°**



PHOTOMETRY

1000 lm, 3000K, Linear Spread **ART2D-10T30LST**
2" Standard Micro Flange Trim, White **X2TSDSWHMF**

Candela Distribution



Luminous Intensity

	0°
0°	1345
5°	1264
10°	1011
15°	704
20°	454
25°	280
30°	168
35°	102
40°	68
45°	50
50°	38
55°	28
60°	20
65°	13
70°	7
75°	4
80°	2
85°	1
90°	0

Zonal Lumen Summary

Zone	Lumens	Luminaire %
0° - 30°	507	76%
0° - 40°	591	88%
0° - 60°	652	97%
0° - 90°	670	100%
0° - 180°	670	100%

Illuminance Chart

Distance from LED	Foot Candles	Diameter
3 ft	149.5 fc	1.7 ft
6 ft	37.4 fc	3.3 ft
9 ft	16.6 fc	5.0 ft
12 ft	9.3 fc	6.7 ft

Input Watts: **12.0 W**
Lumens: **670 lm**
Efficacy: **56 lm/W**
Beam Angle: **31° x 46°**